

Physical Change	Chemical Change
It is a change that occurs without changing the chemical composition of a matter	It is a change of chemical composition of a matter
It is usually reversible	It is usually irreversible
New products are not formed	New products are formed
Some changes occur when heating or or cooling it down	Changes always involve absorption or release of energy
Changes have no effect on the chemical bonds of molecule of a substance	They have a direct effect on chemical bonds of molecules in a substance

Changes

Separation

Filtration

physical separation process that separates solid matter and fluid from a mixture using a filter medium

Hand-picking

method of separating substances in a mixture by hand when the components are clearly visible enough to pick

Sieving

method of using a sieve to distinguish small particles from bigger particles

Evaporation

used for separating a mixture of a non-volatile and a volatile substance

used for separating coloured component from the ink, Salt from water, Sugar from water

Chromatography

used to separate solute that can dissolve in same solvent

used for separating colour components of a dye, drugs from blood

Centrifugation

used for separating dense particles from lighter particles

used to separate cream from Milk, water from wet clothes, centrifugation of blood

Separating Funnel

used to separate to immiscible liquids

used to separate oil from water

Sublimation

used to separate a sublimable component from non-sublimable component

used to separate ammonium chloride/ camphor/ naphthalene and salt

Distillation

used to separate miscible liquids (the boiling point of the liquids must be sufficiently different)

separate acetone and water

types - Simple Distillation & Fractional Distillation

Two or more different type of substance particles & Don't have fixed composition

Homogenous Mixture	Heterogenous Mixture
They have a uniform composition throughout	They have a non-uniform composition
The components of mixture cannot be separated through physical process	The components can be separated through physical process
The component cannot be seen through naked eyes	The components can easily be seen through naked eyes
The mixture is in single-phase throughout Eg: a mixture of water and milk	The substances can be of two different phases, and we may see separate layers of the substances Eg: a mixture of oil in water

Mixture

Tyndall Effect

will not scatter light

Particle size > 1 nm

will scatter light

Particle size > 10 nm

Suspension

when two or more substances are mixed in a non-uniform manner

particle size - greater than 1000 nm

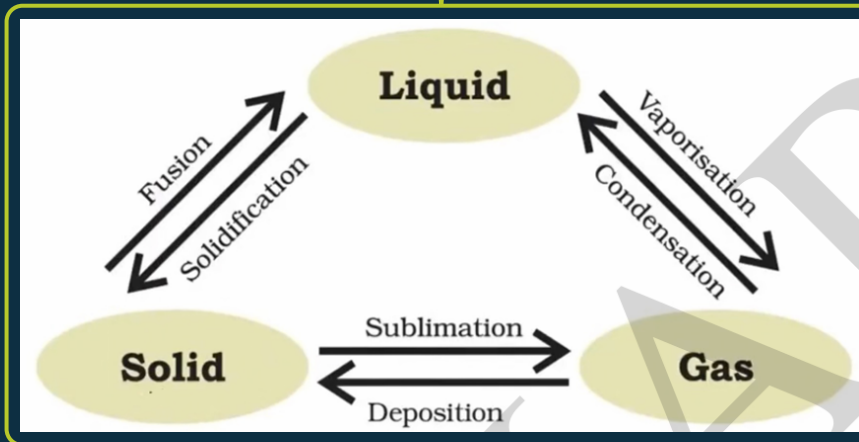
Particle size - 1 nm to 1000 nm

appears to be homogenous but actually is heterogenous

Collidal Solution

Dispersed Phase	Dispersing Medium	Type	Example
Liquid	Gas	Aerosol	Fog, clouds, mist
Solid	Gas	Aerosol	Smoke, automobile exhaust
Gas	Liquid	Foam	Shaving cream
Liquid	Liquid	Emulsion	Milk, face cream
Solid	Liquid	Sol	Milk of magnesia, mud
Gas	Solid	Foam	Foam, rubber, sponge
Liquid	Solid	Gel	Jelly, cheese, butter
Solid	Solid	Solid Sol	Coloured gemstone, milky glass

Change in Matter



States of matter

	space	Kinetic Energy	Particle Movement	Force of attraction
Solid	Low	Low	Low	High
Liquid	Medium	Medium	Medium	Medium
Gas	High	High	High	Low

Plasma

4th State of Matter

made up of free-moving electrons, positive ions & neutral particles

good conductor of electricity & It's glows

5th state of Matter

formed when a gas of bosons at very low densities is cooled to temperatures very close to absolute zero (0 K/ -273.15 °C/ -459.67 °F)

Bose-Einstein condensate (BEC) - name before Dr. Satyendra Nath Bose & Einstein

1924 - Dr. S N Bose published research paper
1925 - sent it to Einstein

1995 - Eric Cornell & Carl Wieman experimentally produced 1st BEC condensate in a rare field gas of Rubidium atoms
1995 - Wolfgang Ketterle created BEC condensate with Sodium atoms
2001 - All three received Noble Prize in Physics

Boson

Pure Substance

it is made up of only one type of particle & has uniform composition & properties

eg: Sugar crystal, iron piece, salt, etc.

Element

Can't be broken into simple substances

Compound

Can be broken into simple substances

has two components - Solute & Solvent

Partical size - less than 1 nm

Saturated solution - Not much solute can be dissolved

Concentrated Solution - Solute > Solvent

Dilute Solution - Solvent > Solute

Concentration of Solution = $\frac{\text{Amount of Solute}}{\text{Amount of Solution}}$

Mass by % = $\frac{\text{Mass of Solute} \times 100}{\text{Mass of Solution}}$

Liquid into liquid - Water and Ink
Solid into Solid - Alloys
Gas into Gas - Air
Solid into a liquid - Sugar and Water
Solid into gas - Hydrogen and metal
Liquid into gas - Carbon Dioxide and Water

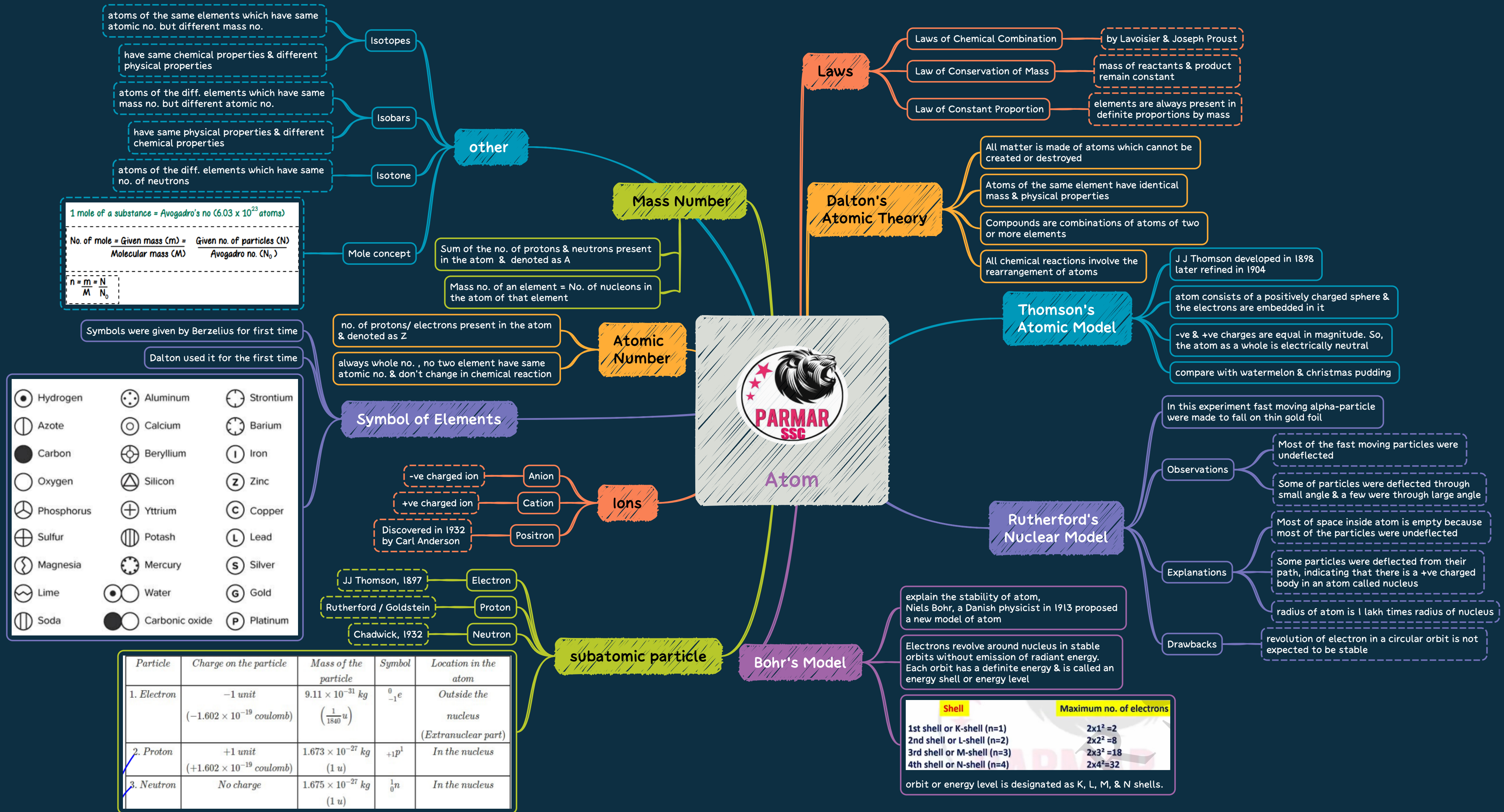
type of Solid

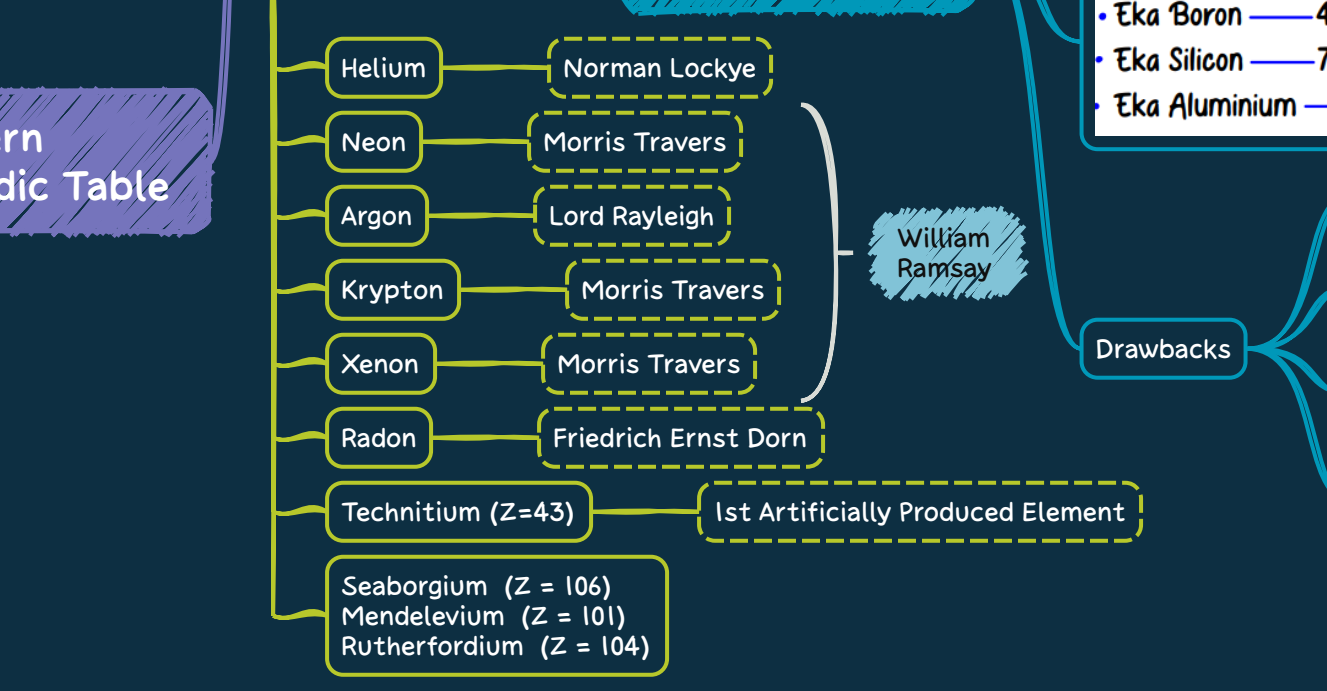
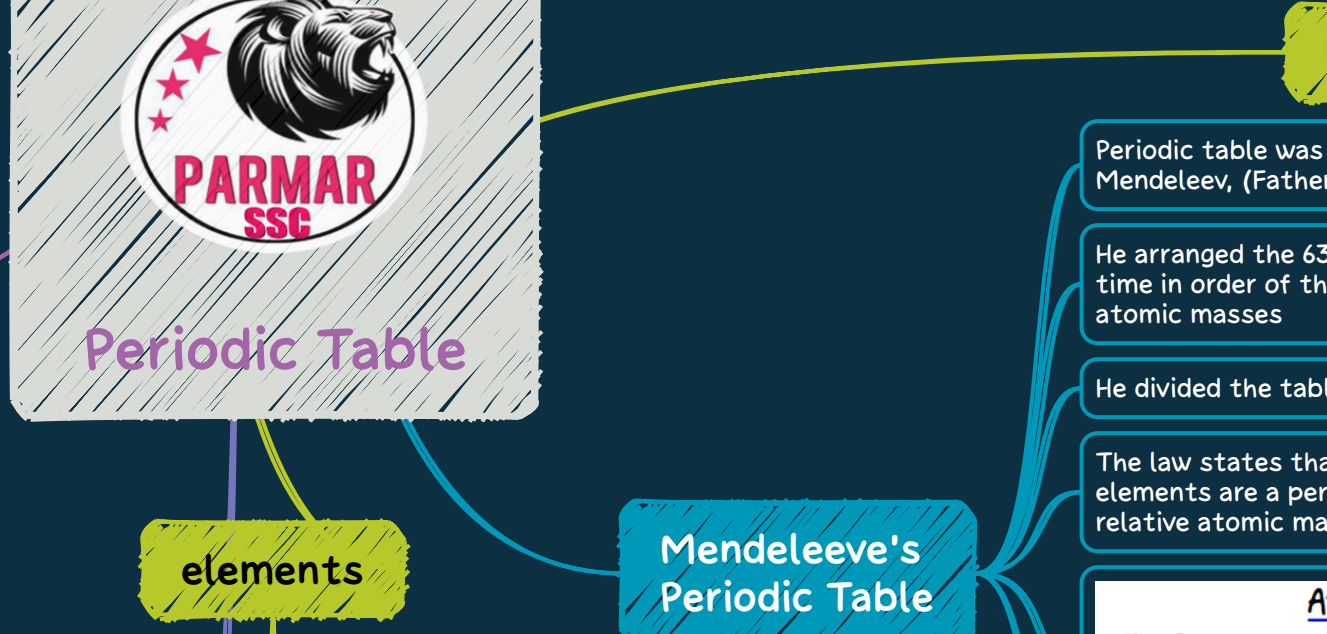
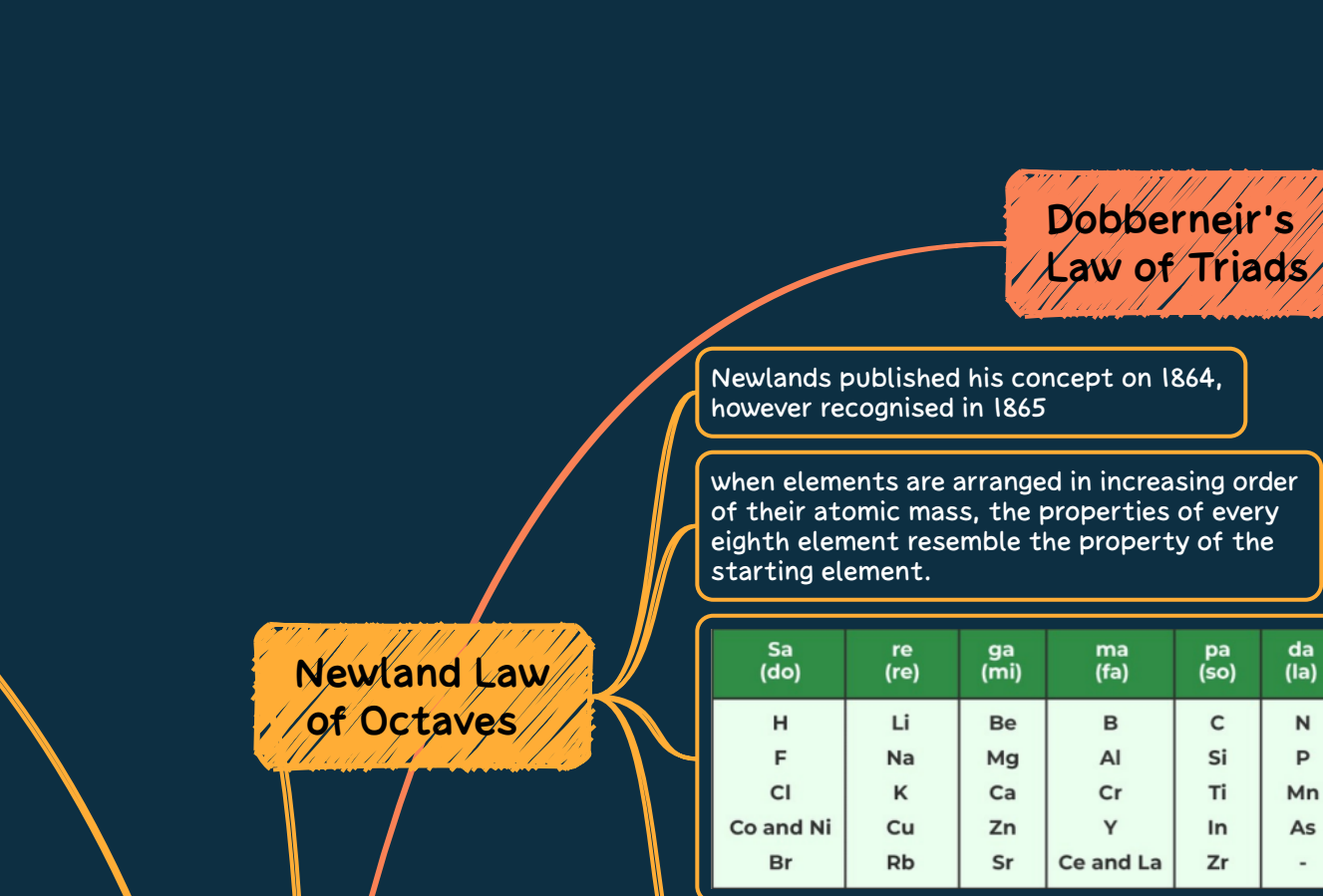
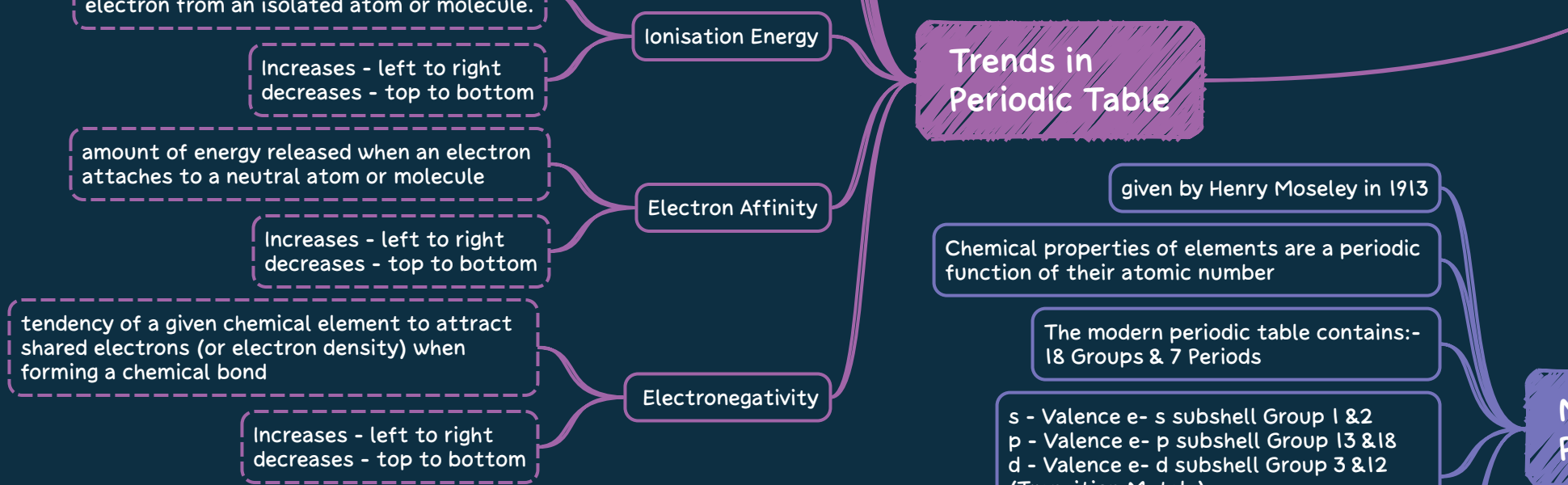
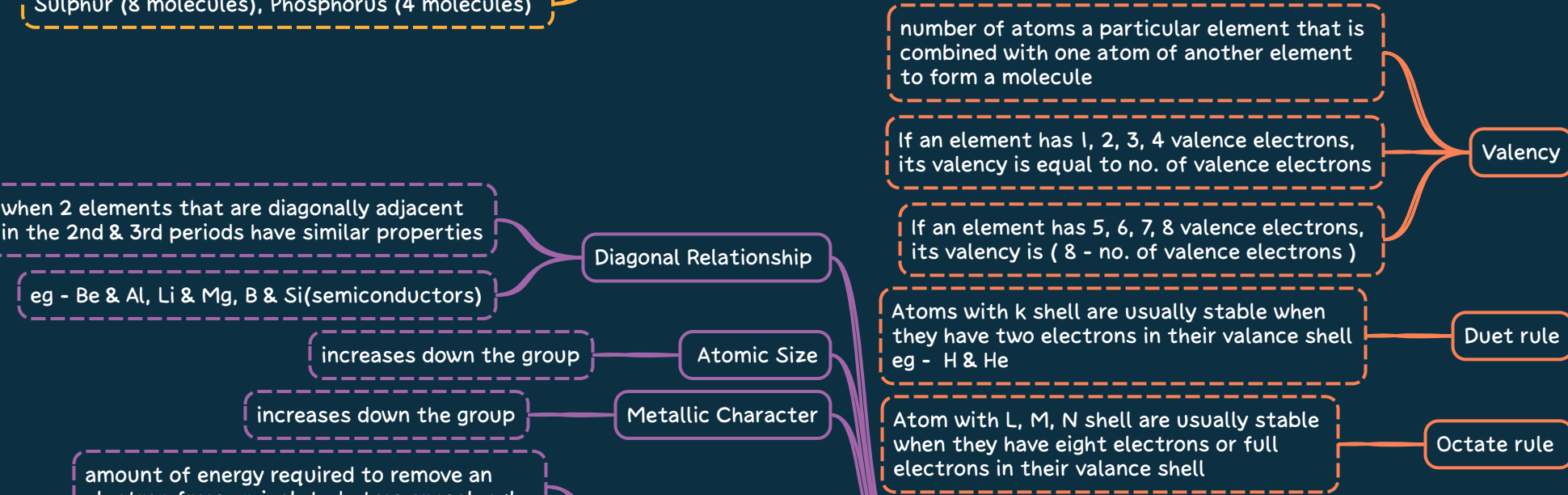
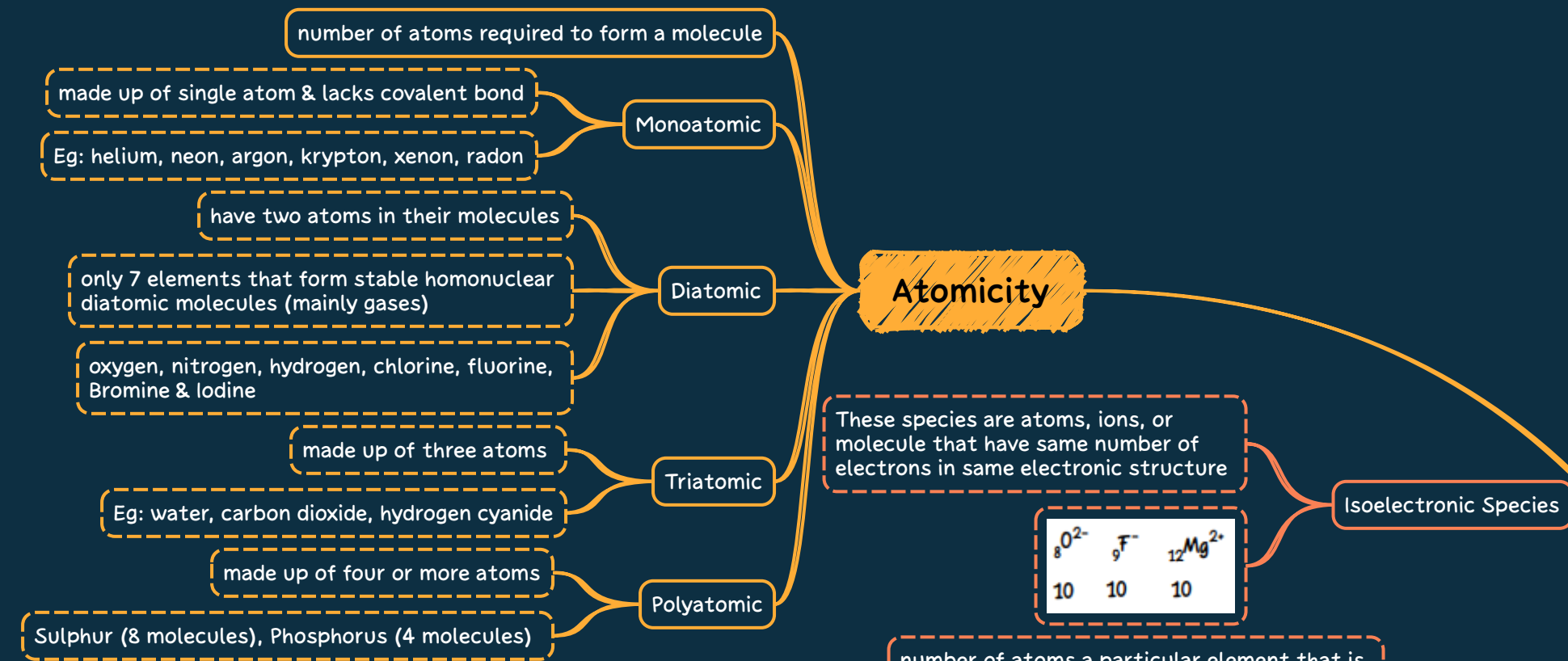
Amorphous Solid

- They are isotropic in nature
- They have irregular shape
- They are pseudo, solids are super cooled liquids
- They have irregular arrangement of constituent particles
- They do not have sharp melting point
- Eg: Glass, rubber, plastic, etc.

Crystalline Solid

- They are anisotropic in nature
- They have definite characteristics shape
- They are true solids
- They have regular arrangement of constituent particles
- They have sharp melting point
- Eg: Copper, Silver







Reaction

Exothermic

A reaction that releases energy from the system in the form of heat

Endothermic

reaction in which the system absorbs energy from its surrounding in the form of heat

Day-to-Day Life Examples

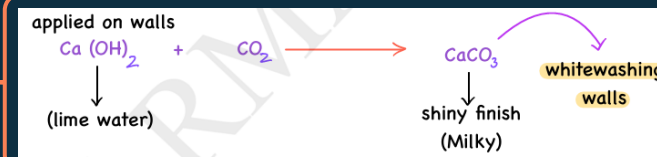
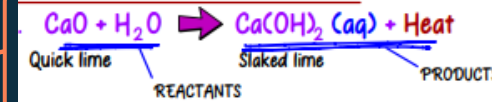
Corrosion



When fats gain oxygen, it oxidises & thus causes rancid smell.

Rancidity

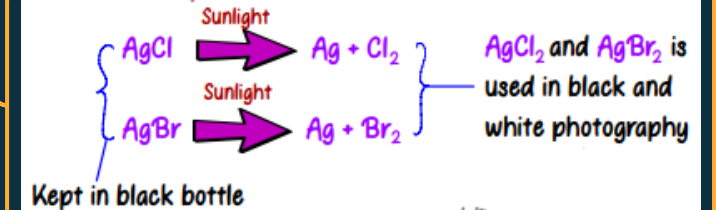
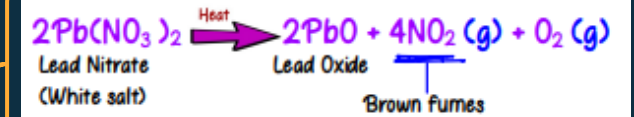
Combination Reactions



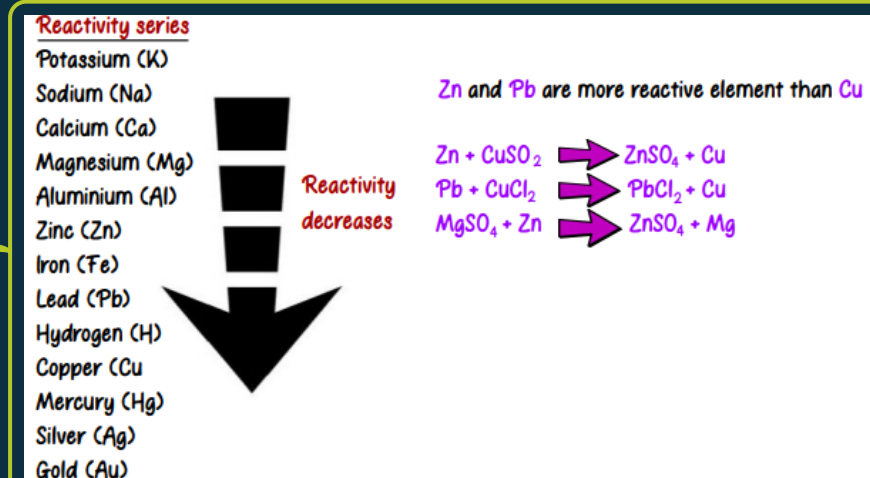
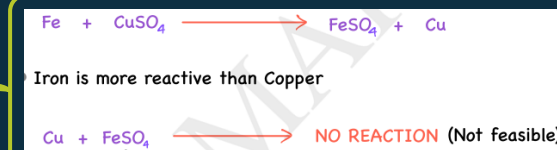
Most of the combination reactions are exothermic in nature

Decomposition Reactions

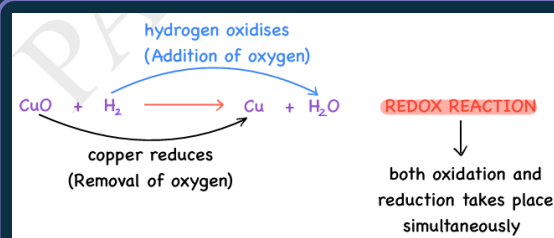
usually ENDOTHERMIC



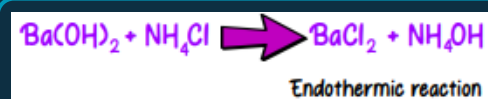
Displacement Reactions



Oxidation & Reduction



Double Displacement Reaction



Brass	Cu (80%) + Zn (20%)
Bronze	Cu (88%) + Sn (12%)
Solder	Sn + Pb
Bell Metal	Cu + Sn
Gun Metal	Cu + Sn + Zn
Amalgam	Hg + other metals
Stainless Steel	Fe + Ni + Cr + Carbon
Constantan	Cu + Ni
Manganin	Cu + Ni + Mn
Nichrome	Ni + Cr + Mn + Fe
German Silver	Cu + Ni + Zn
White Gold	Au (75%) + Ni + Pd
Magnallium	Al (95%) + Mg
Duralumin	Al (95%) + Mg + Mn + C

Alloy

Occurrence of Metals

1. Aluminium:- Bauxite; Cryolite
2. Iron:- Magnetite, Haematite, Limonite, Siderite
3. Zinc:- Zinc Blende/Sphalerite; Calamine; Zincite (Zinc)
4. Mercury:- Cinnabar
5. Thorium:- Monazite
6. Sodium:- Borax; Chile and Indian saltpeter (KNO_3); Trona
7. Lead:- Galena
8. Magnesium:- Epsom salt ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$)
9. Tin:- Cassiterite
10. Uranium:- Pitchblende (it is made up of Uranium Oxide (UO_2) and variable proportions of U_3O_8)

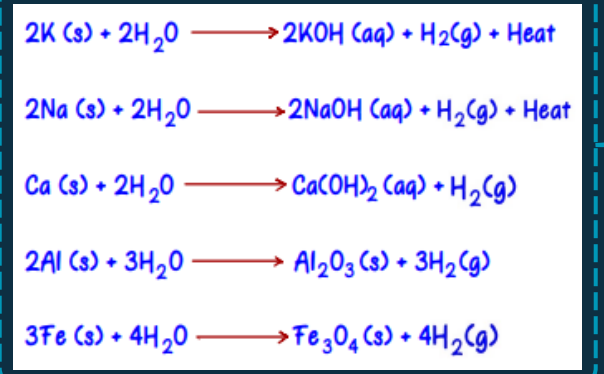
Ore has: Metal + Gangue (impurities)

Metal + Oxygen $\rightarrow$ Metal Oxide (basic)

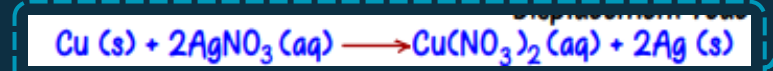
Amphoteric Oxides are both basic & acidic eg: SnO_2 , BeO , ZnO , Al_2O_3 , PbO_2 etc

Most Metal Oxides are insoluble in Water exception - Sodium Oxide & Potassium Oxide

Metal + Water $\rightarrow$ Metal Oxide + Hydrogen
Metal + Water $\rightarrow$ Metal Hydroxide

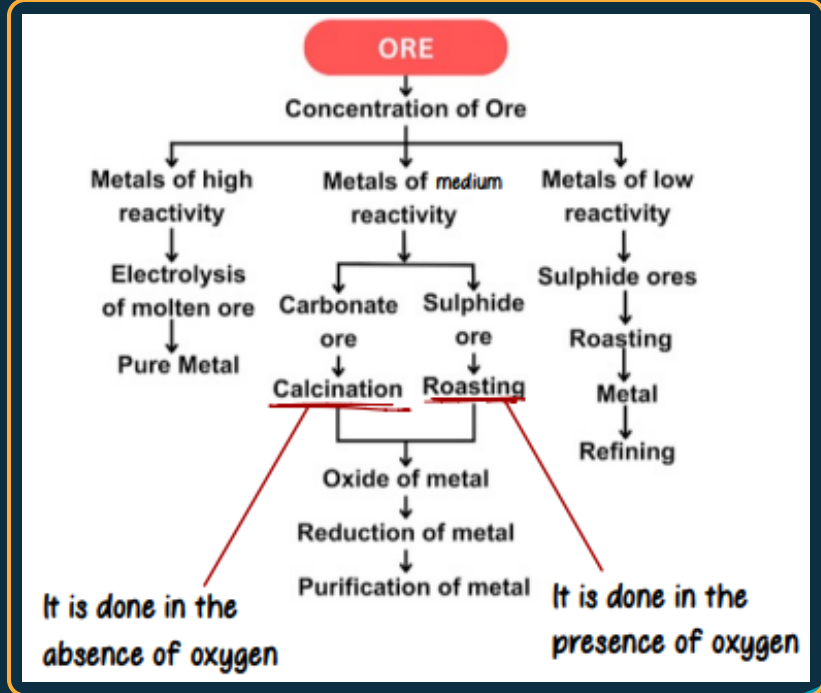
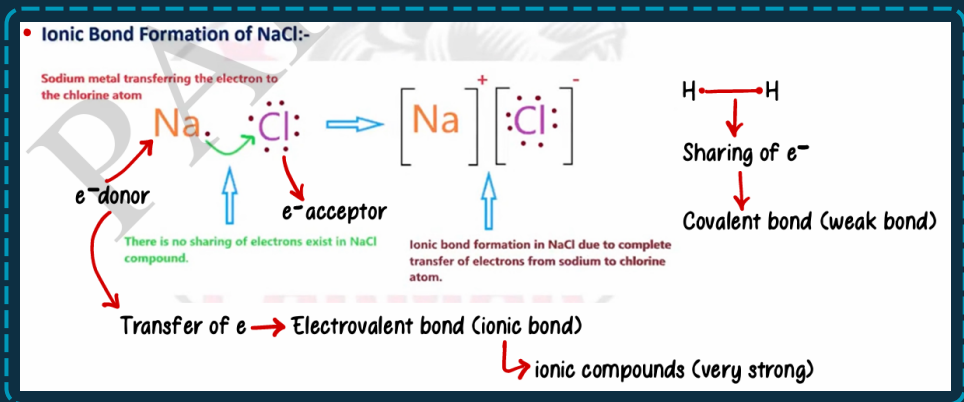


Metal + Dilute Acid $\rightarrow$ Salt + Hydrogen



Metal(1) + Metal(2) Salt $\rightarrow$ Metal(1) Salt + Metal(2)

Non-metal + Oxygen $\rightarrow$ Non-metal oxide (acidic)



Chemical properties

Non-metal

Potassium	K	Most reactive metal	Reacts vigorously with cold water (highly Exothermic)
Sodium	Na		
Barium	Ba		
Calcium	Ca		Reacts vigorously with hot water
Magnesium	Mg		Reacts vigorously with steam
Aluminium	Al		
Zinc	Zn		
Iron	Fe		
Nickel	Ni		
Tin	Sn		
Lead	Pb		
Hydrogen	(H)		
Copper	Cu		
Mercury	Hg		
Silver	Ag		
Gold	Au		
Platinum	Pt	Least reactive metal	No reaction with any type of water

Decreasing chemical reactivity



Metals & Non-metals

Physical State

- Metal
 - Generally exists in solid state
 - Except - Hg, Ga, Cs
- Non-metal
 - exists in solid & gaseous state
 - Except - Br (liquid)

Lusture

- Metal
 - They have shining surface
 - Except - lead
- Non-metal
 - They lack lusture
 - Except - Iodine, Diamond, & Graphite

Hardness

- Metal
 - Are hard in nature
 - Except - Na, K, Cs, Hg, Ga, Zn
- Non-metal
 - Are soft in nature
 - Except - Diamond

Malleability

- Metal
 - Beaten into sheets
 - Except - Hg, Na, K, Zn
- Non-metal
 - They lack ductility

Ductility

- Metal
 - Drawn into wire
 - Except - Hg, Na, K, Zn
- Non-metal
 - They are not malleable

Melting & Boiling point

- Metal
 - high melting and boiling point
 - Except - Hg
- Non-metal
 - low melting and boiling point
 - Except - Diamond, Graphite, Si, C, B

Density

- Metal
 - They have high density
 - Except - Na and K
- Non-metal
 - Have low density
 - Except - Diamond

Brittleness

- Metal
 - They are hard
 - Except - Zn
- Non-metal
 - Are brittle
 - Except - gases

Sonorous

- Metal are sonorous
- Non-metal are not sonorous

Reactivity Series

Thermal & electrical conductivity

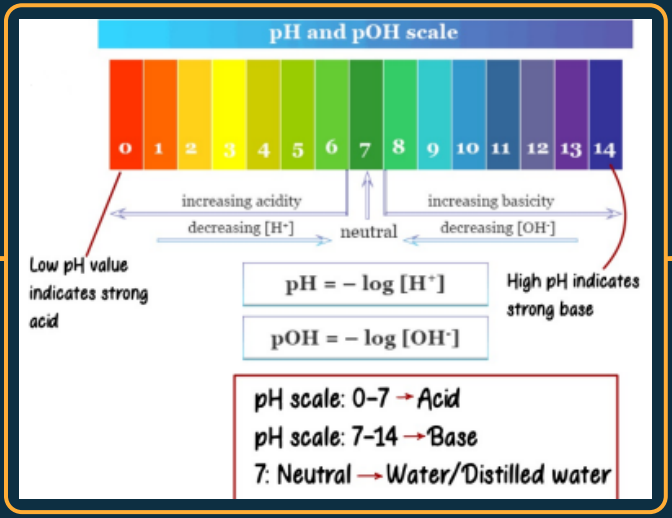
- Metal
 - Are good conductor of heat & electricity
 - Except - lead & Hg
 - Cationic Property
- Non-metal
 - Bad conductor of heat & electricity
 - Except - Graphite
 - Anionic Property

Metalloids

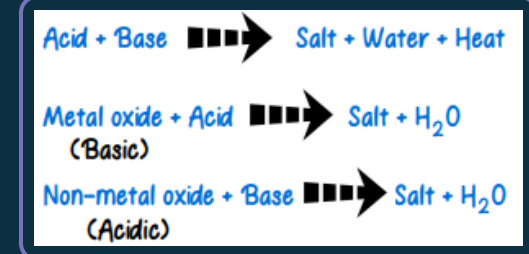
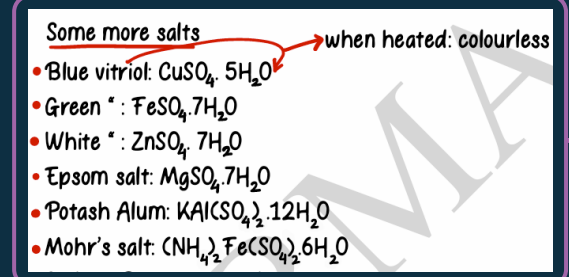
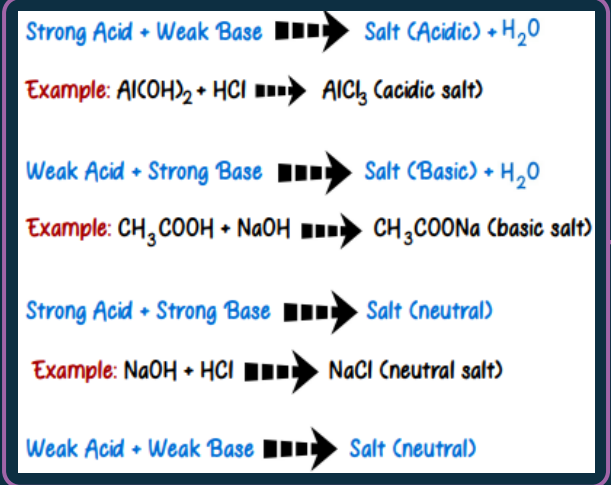
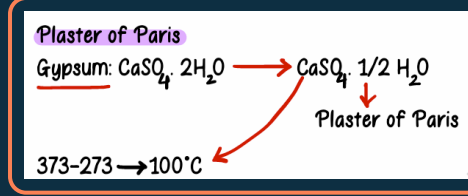
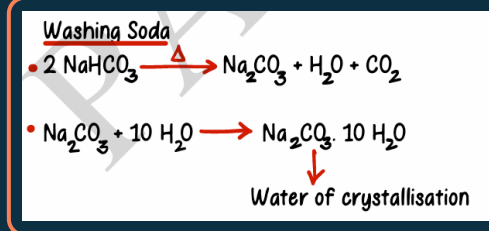
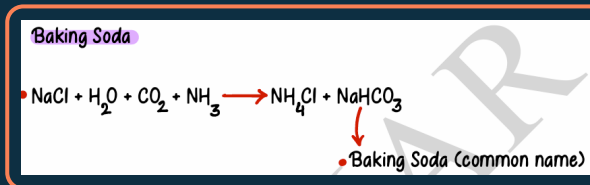
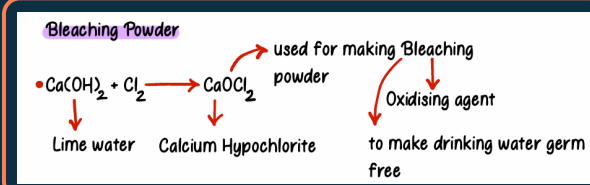
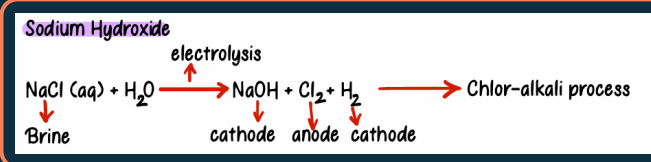
- having Intermediate properties between Metal & Non - Metals
- Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium & Polonium

Our body works in a pH of 7-7.8

- Gastric juice: 1.6
- Lemon: 2
- Milk: 6.4
- Blood: 7.4 (slightly alkaline)
- Saliva: 6.5
- Milk of Magnesia: 10
- NaOH: 14



Imp. reaction



pH Scale

Acid

Base

Acid, Base & Salt

Sour substance, Corrosive in nature

Organic Acid

Acetic Acid	Vinegar
Citric Acid	Citrus Fruits (Lemon, orange)
Tartaric Acid	Tamarind, Grapes, Kaccha Mango
Oxalic Acid	Tomato, Spinach
Lactic Acid	Curd
Ascorbic Acid	Citrus Fruits, Amla
Methanoic/Formic Acid	Ant Sting, Nettle Sting
Butyric Acid	Rancid Butter
Stearic Acid	Oil/ Fats
Maleic acid	Apple, Banana

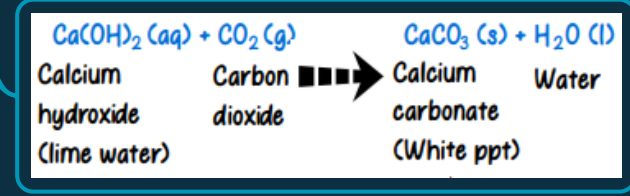
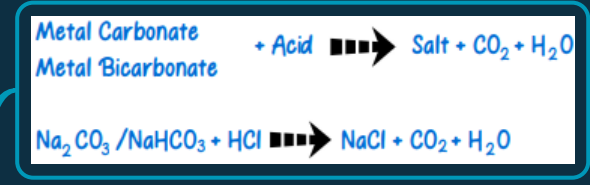
Mineral Acid

Sulphuric Acid	Oil of Vitriol
Nitric Acid	Explosives
Hydrochloric Acid	Gastric Juice
Boric Acid	Antiseptic
Carbonic Acid	Cold Drinks
Phosphoric Acid	Fertilizers

Bitter substance

Base

Calcium Hydroxide	Lime Water
Ammonium Hydroxide	Window Cleaner
Sodium / Potassium Hydroxide	Soap
Magnesium Hydroxide	Milk of Magnesia



Indicators

indicates if anything is an Acid/Base

Litmus paper

- Belongs to Thallophyta
- natural indicator
- Blue to Red in Acid
- Red to Blue in Base

Turmeric

- Acid: Remains yellow
- Blue: Turn to reddish brown

Phenolphthalein

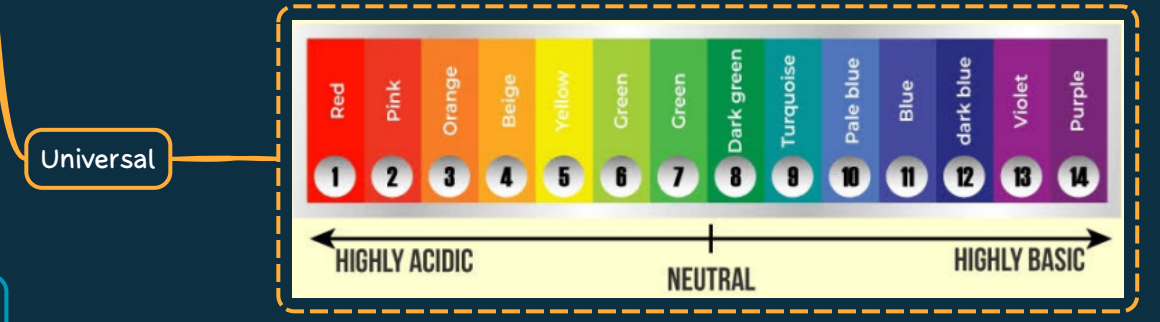
- Acid: Colourless
- Base: Pink

Methyl Orange

- Acid: Red
- Base: Yellow

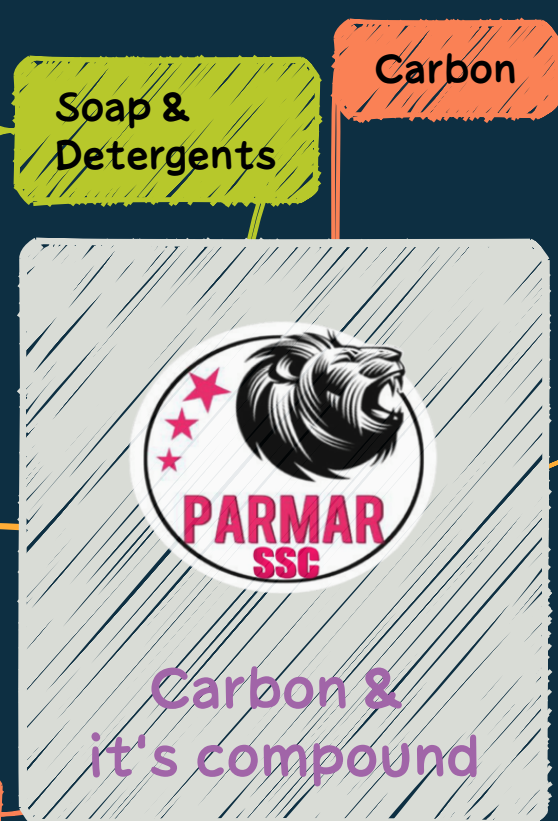
Olfactory

- Onion**
 - Acid: remains smell
 - Base: loses it's smell
- Vanila Extract**
 - Acid: remains smell
 - Base: loses it's smell
- Clove Oil**
 - Acid: remains smell
 - Base: loses it's smell



Chemical Properties

Reaction

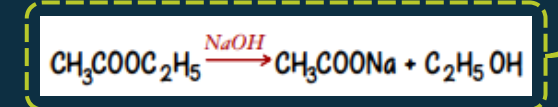


Carbon

- Atmosphere - 0.036%
- Earth crust - 0.025%
- Atomic no - 6 (2, 4)
- Allotropes
 - Graphite - 1C --> 3C
 - Diamond - 1C --> 4C
 - Buckmister Fullerene - C 60

Soap & Detergents

Saponification Reaction



Used in preparation of soaps on treating ester with Sodium Hydroxide

Soap

does not work in hard water & Acidic condition

Detergents

Sodium salt of sulphonic acids or ammonium salts with Chloride, Bromide ions etc.

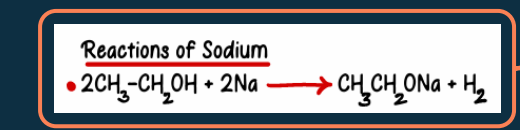
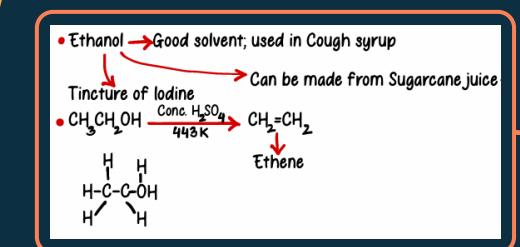
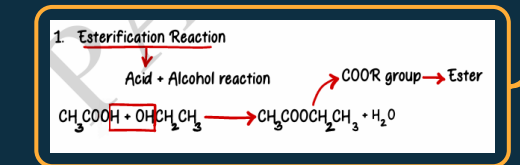
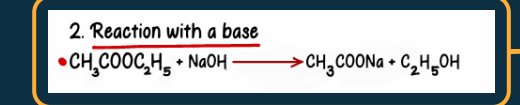
Work in both hard and soft water

Dissolve in water faster than soap

Ethanoic Acid

Commonly known as Acetic acid

Also known as Glacial Acetic Acid as Melting Point is 290 K



Ethanol

Nomenclature

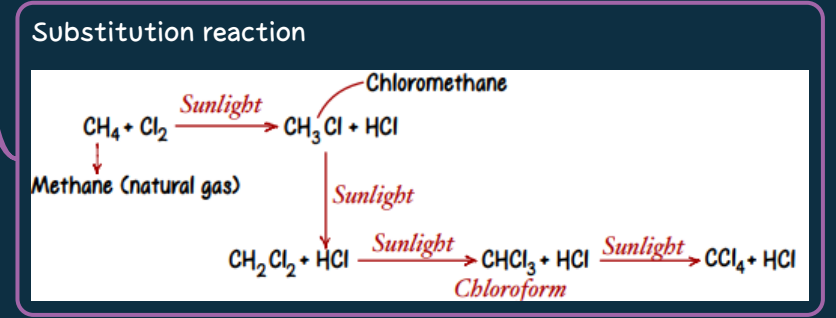
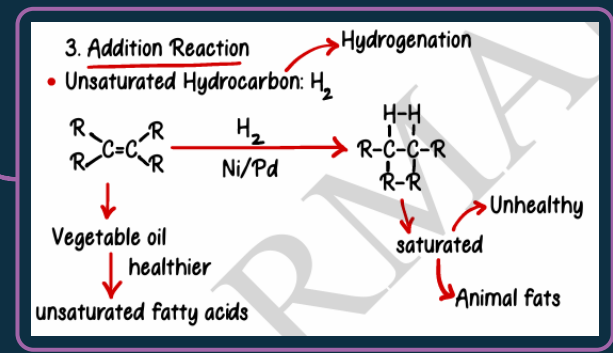
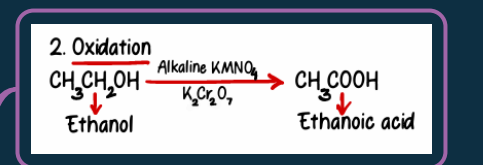
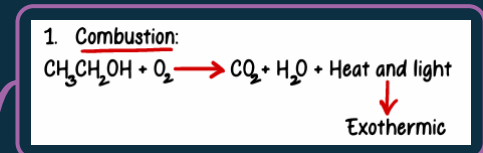
Chemical Properties

Versatile Nature

Catenation - unique ability to form bonds with other carbon atom (Large molecule)

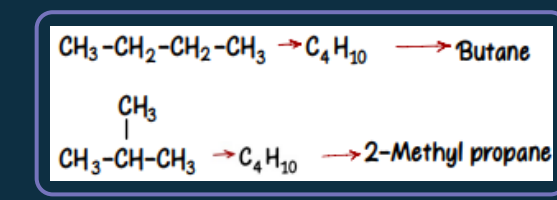
- ane: C-C single bond Saturated compounds
- ene: C=C Unsaturated compounds
- yne: C≡C

- Alkane: $\text{C}_n\text{H}_{2n+2}$
- Alkene: C_nH_{2n}
- Alkyne: $\text{C}_n\text{H}_{2n-2}$



Isomer

have same chemical formula but diff. structure



no. of isomers: $2^{(n-4)} + 1$

Homologous series

Consist of one functional group

Differs by -CH unit and mass by 14 units.

Chemical properties remains same

Alkanes	Alkynes	Alcohols
CH_4	C_2H_2	CH_3OH
C_2H_6	C_3H_4	$\text{C}_2\text{H}_5\text{OH}$
C_3H_8	C_4H_6	$\text{C}_3\text{H}_7\text{OH}$
C_4H_{10}	C_5H_8	$\text{C}_4\text{H}_9\text{OHCH}_2$

Class of compounds	Functional group structure	IUPAC group prefix	IUPAC group suffix	Example
Alkanes	-	-	-ane	Butane, $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$
Alkenes	>C=C<	-	-ene	But-1-ene, $\text{CH}_2=\text{CHCH}_2\text{CH}_3$
Alkynes	$\text{-C}\equiv\text{C-}$	-	-yne	But-1-yne, $\text{CH}\equiv\text{CCH}_2\text{CH}_3$
Arenes	-	-	-	Benzene,
Halides	-X (X=F, Cl, Br, I)	halo-	-	1-Bromobutane, $\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{Br}$
Alcohols	-OH	hydroxy-	-ol	Butan-2-ol, $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$
Aldehydes	-CHO	formyl, or oxo	-al	Butanal, $\text{CH}_3(\text{CH}_2)_2\text{CHO}$
Ketones	>C=O	oxo-	-one	Butan-2-one, $\text{CH}_3\text{CH}_2\text{COCH}_3$
Nitriles	$\text{-C}\equiv\text{N}$	cyano	nitrile	Pentanenitrile, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CN}$
Ethers	-R-O-R-	alkoxy-	-	Ethoxyethane, $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
Carboxylic acids	-COOH	carboxy	-oic acid	Butanoic acid, $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{H}$
Carboxylic salts	-COO-	-	-ate	Sodium butanoate, $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{Na}$
Esters	-COOR	alkoxycarbonyl	-oate	Methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$